PROBLEMA $\longrightarrow$ $x_{[n]} = 4 + 3\sin\left[\frac{\pi}{2}\cdot n\right] \xrightarrow[\text{DFT}]{N=8 \text{ MUESTRAS}} X_{[k]}$

$$X_{[k]} \triangleq \sum_{n=0}^{N-1} x_{[n]}\, e^{-j\frac{2\pi}{N}kn}$$

$$X_{[k]} = 4\sum_{n=0}^{N-1} e^{-j\frac{2\pi}{N}kn} + \frac{3}{2j}\sum_{n=0}^{N-1}\left(e^{j\frac{\pi}{2}n} - e^{-j\frac{\pi}{2}n}\right) e^{-j\frac{\pi}{4}kn}$$

$$X_{[k]} = 4\sum_{n=0}^{N-1} e^{-j\frac{2\pi}{N}kn} + \frac{3}{2j}\sum_{n=0}^{N-1} e^{j\frac{\pi}{2}n} - e^{-j\frac{\pi}{4}nk} - \frac{3}{2j}\sum_{n=0}^{N-1} e^{-j\frac{\pi}{2}n} - e^{-j\frac{\pi}{4}nk}$$

- REMPLAZO POR LA APROXIMACIÓN DE LA SERIE GEOMÉTRICA

$$X_{[k]} = 4\cdot \underbrace{\frac{1-\overbrace{\left(e^{-\frac{j2\pi k}{N}}\right)^{N}}^{=0\ \forall k}}{1-e^{-\frac{j2\pi k}{N}}}}_{=1,\ k=0} + \frac{3}{2j}\underbrace{\frac{1-e^{j\pi\left(\frac{1}{2}-\frac{1}{4}k\right)8}}{1-e^{j\pi\left(\frac{1}{2}-\frac{1}{4}k\right)}}}_{=0\ \forall k\neq 2} - \frac{3}{2j}\frac{1-e^{j\pi\left(-\frac{1}{2}-\frac{1}{4}k\right)8}}{1-e^{j\pi\left(-\frac{1}{2}-\frac{1}{4}k\right)}}$$

$$X_{[k]} = 4\cdot\delta_{[k]} + \frac{3}{2j}\,\delta_{[k-2]} - \frac{3}{2j}\,\delta_{[k-6]}$$